

SERVERLESS COMPUTING WITH AWS LAMBDA

❖ What is Serverless Computing?

- Serverless computing is a way to **run applications** without **managing servers** yourself.
- While servers are still involved, their provisioning, scaling, and maintenance are handled by the cloud provider.

❖ What is AWS Lambda?

- AWS Lambda is Amazon's serverless computing service - it's the most popular way to run serverless functions.
- **What Lambda Does:** Lambda lets you **run code** without **provisioning or managing servers**. You just upload your function, and AWS handles everything else - servers, scaling, maintenance, and monitoring.

✓ How It Works:

1. You write a function (in Python, Node.js, Java, C#, etc.)
2. Upload it to Lambda
3. Set up triggers (what makes your function run)
4. Lambda automatically executes your function when triggered
5. You pay only for execution time

❖ Benefits of AWS Lambda

1. No Server Management
2. Automatic Scaling
3. Pay-Per-Use Pricing
4. High Availability
5. Built-in Monitoring
6. Multiple Language Support

❖ Real-Time Use Cases of AWS Lambda

1. Image/Video Processing
2. Real-Time Data Processing
3. Web APIs and Microservices
4. Chatbots and Voice Assistants
5. File Processing Automation
6. Scheduled Tasks

❖ Advantages of Serverless Computing

1. Automatic Scaling
2. Cost Savings
3. Improved Productivity
4. High Availability

[GENERAL INTERVIEW QUESTIONS]

◆ 1. What is AWS Lambda and how does it differ from EC2?

- Lambda is a serverless compute service that runs code without provisioning or managing servers, unlike EC2 which requires full server management.

◆ 2. What triggers can invoke a Lambda function?

- Common triggers include S3 events, API Gateway, DynamoDB Streams, CloudWatch Events, and SNS.

◆ 3. What is an event object in Lambda?

- An event is a JSON-formatted input passed to the Lambda function, containing trigger data like S3 file names or API payloads.

◆ 4. How does AWS Lambda handle scaling?

- Lambda automatically scales by running multiple instances of the function in parallel — each triggered by an event.

◆ 5. What is a Lambda Execution Role?

- An IAM role that gives Lambda permission to access AWS services (e.g., S3 read, DynamoDB write).

◆ 6. How do you deploy dependencies in Lambda?

- Package your code with dependencies as a .zip file or use Lambda layers for sharing libraries across functions.

◆ 7. What is the difference between synchronous and asynchronous invocation in Lambda?

- **Sync:** Waits for response (e.g., API Gateway)
- **Async:** Doesn't wait (e.g., S3 events, SNS)

◆ 8. How would you run a Lambda function on a schedule?

- Use CloudWatch Events / EventBridge with a cron or rate expression to trigger the Lambda.

◆ 9. Can Lambda functions interact with private VPC resources?

- Yes. You must configure VPC settings (subnets + security groups) in the Lambda configuration.

◆ 10. What are the limitations of AWS Lambda?

- Max runtime: 15 mins
- Max memory: 10,240 MB
- Max package size: 50 MB (inline), 250 MB (with S3)

◆ 11. What tools can you use to deploy Lambda functions?

- AWS CLI
- AWS SAM (Serverless Application Model)
- Terraform/CloudFormation
- Console
- CI/CD tools (e.g., GitHub Actions, Code Pipeline)